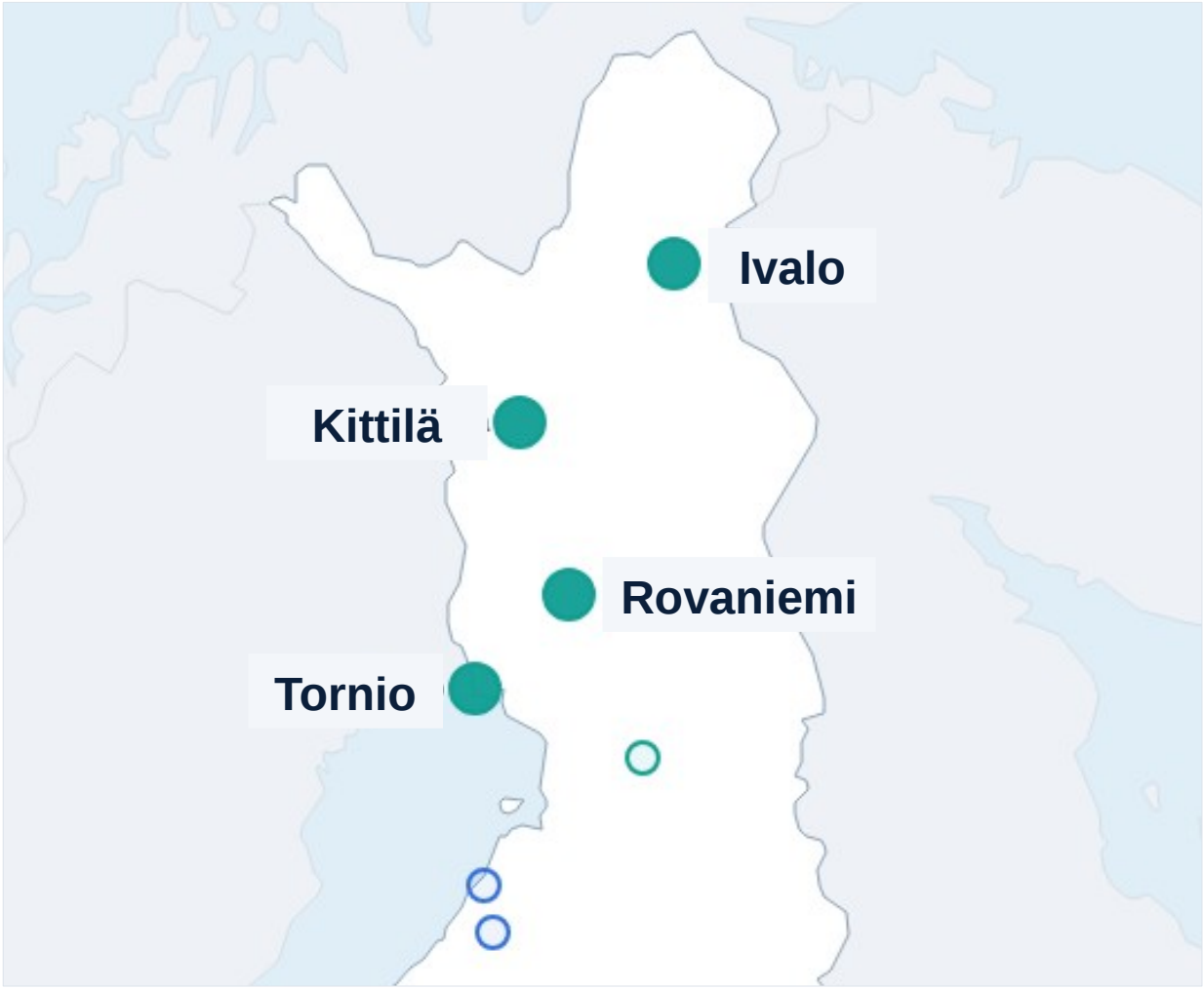
In May 2005, Kittilä and Ivalo flooded together.

Lapland's snow had warned us two months earlier.



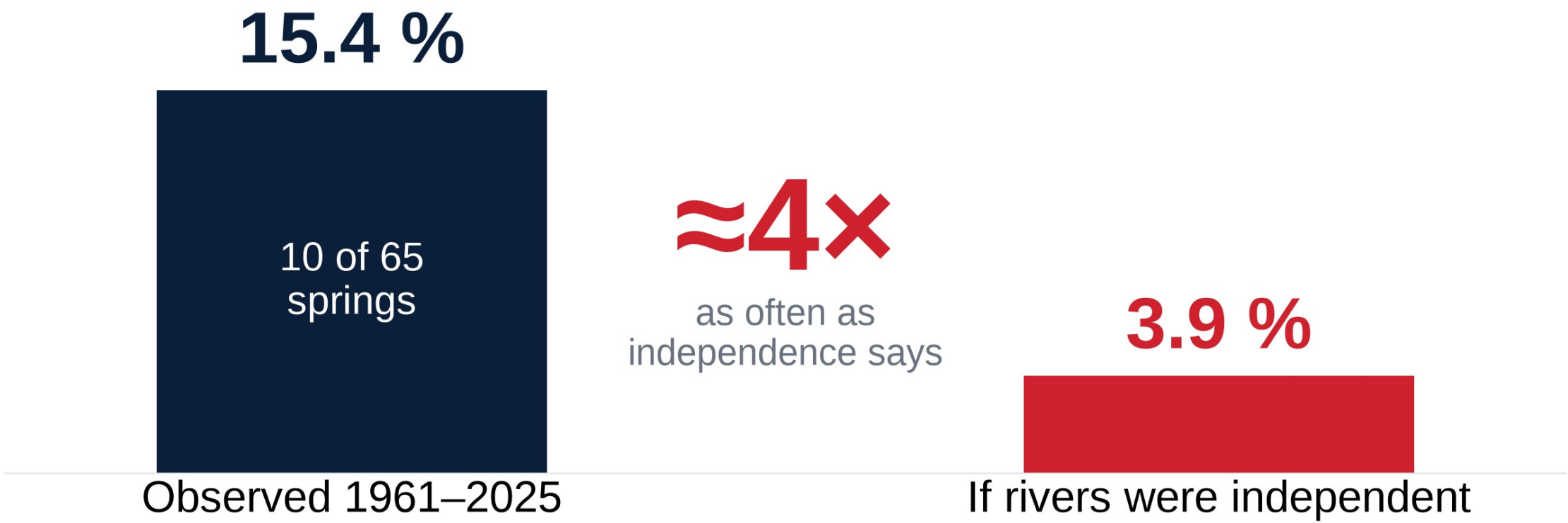
Tulva app: spring 2005 replay

Source: SYKE discharge 1961–2025 and 1 March snow (SWE); Tulva back-test (out/backtest.json)

THE GAP

Finland's rivers flood together, not one at a time

Springs when ≥4 of 14 river areas pass their 1-in-10 flood level



Cover is bought in autumn. The snow signal comes in March.

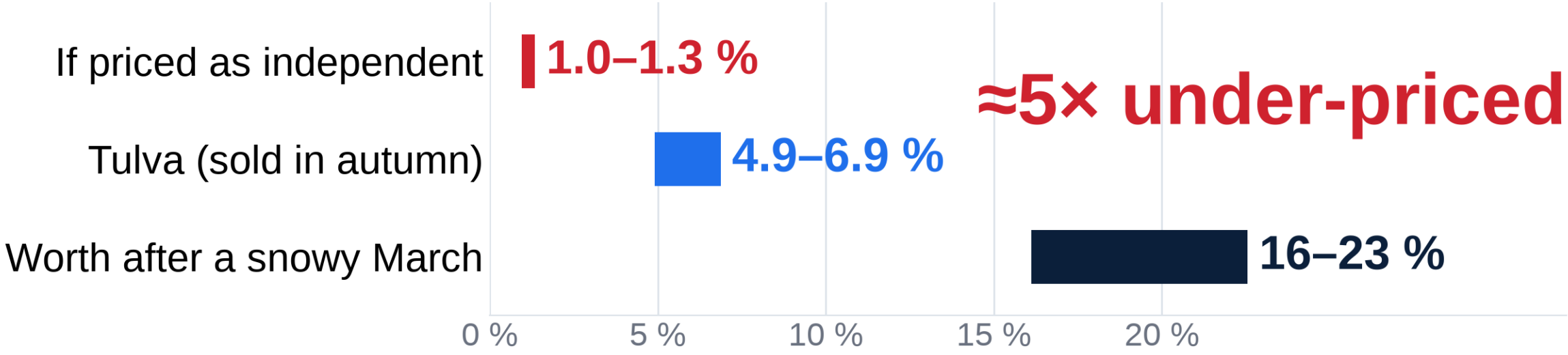
Source: SYKE 1961–2025 (+ SMHI Torne); independence = same gauge-years, no co-flooding

THE PRODUCT

Tulva Index: sold in autumn, updated in March

Pays if ≥2 of 5 northern rivers pass their 1-in-50 flood level

MARKET PRICE, % OF THE LIMIT PER YEAR



Example: €3M limit ≈ €150–210k / year

Sources: SYKE 1961–2025, Tulva model; market 2.5–3.5× expected payout (AM Best cat bonds)

Tulva

Joint risk | Snow outlook | Index cover | How it works | Under the hood | Year replay

1 · JOINT RISK

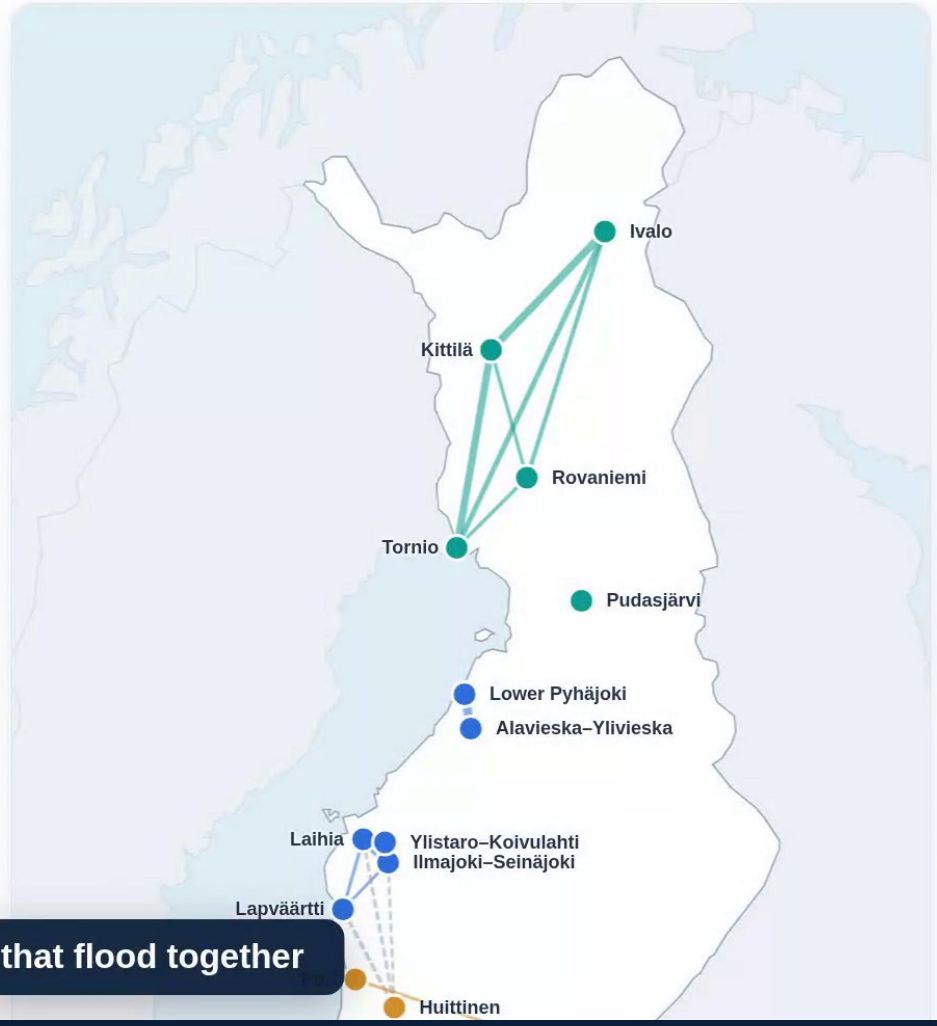
Finland's spring floods don't arrive one at a time.

The 14 river flood-risk areas fall into three clusters that share a snowpack. Lines join areas whose spring peaks move together (Spearman ρ ≥ 0.6, 1961–2025).

2.8×
more joint 1-in-10 floods within a cluster than independence predicts EXACT

10 of 65
years with ≥4 of 14 areas over their 1-in-10 level EXACT

- North · 5 areas · 2.9× joint floods Pudasjärvi, Ivalo, Kittilä, Rovaniemi, Tornio
- Ostrobothnia · 6 areas · 3.1× joint floods Lapväärtti, Ilmajoki–Seinäjoki, Laihia, Ylistaro–Koivulahti, Alavieska–Ylivieska, Lower Pyhäjoki
- Satakunta–SE · 3 areas · 1.3× joint floods Pori, Huittinen, Lower Kymijoki



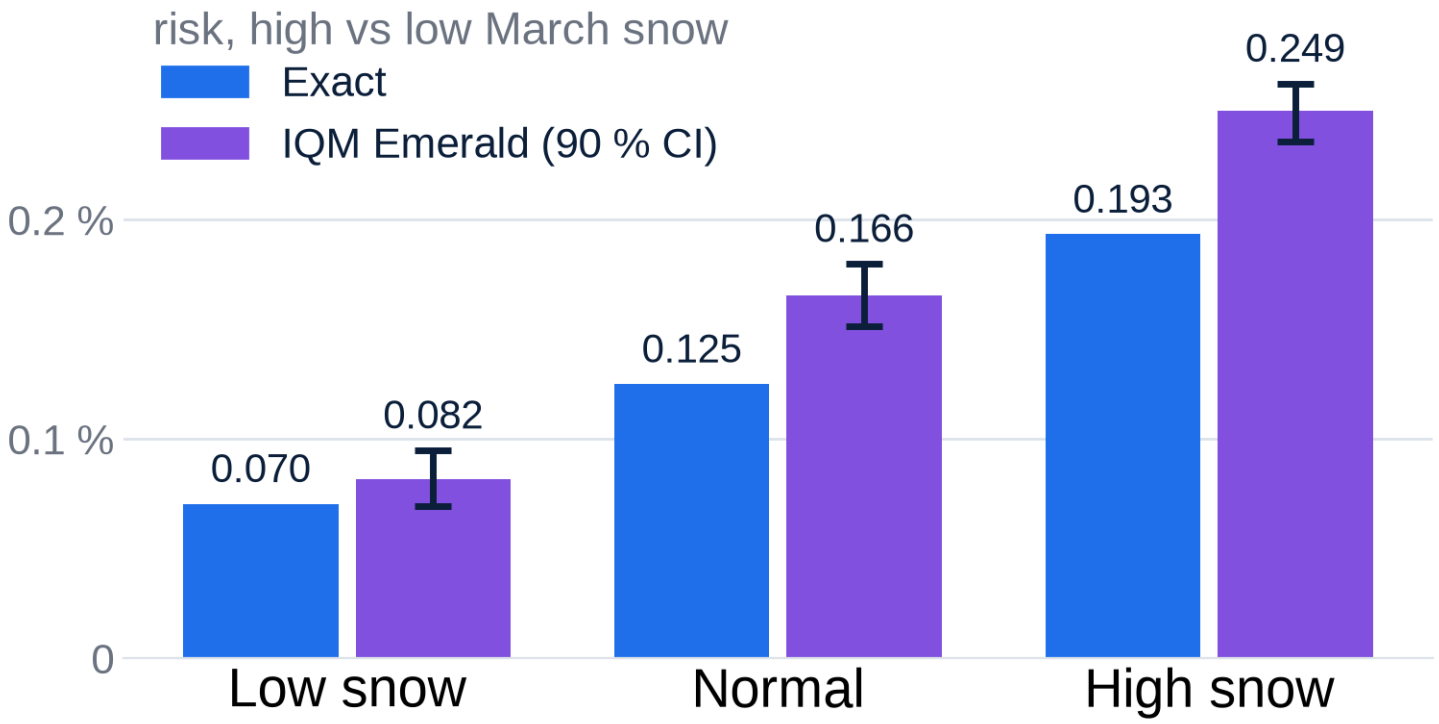
Live app: joint-flood map → March snow → quantum chip → 2005 replay

QUANTUM, IN PRACTICE

It runs on real quantum chips today

IQM Emerald: Kittilä and Ivalo both flood

3.05× on chip vs 2.75× exact



ALL 14 AREAS, ONE CHIP

18 qubits, 44 two-qubit gates on IQM Emerald (sampling)

AMPLITUDE ESTIMATION

Kittilä + Ivalo, 1-in-50

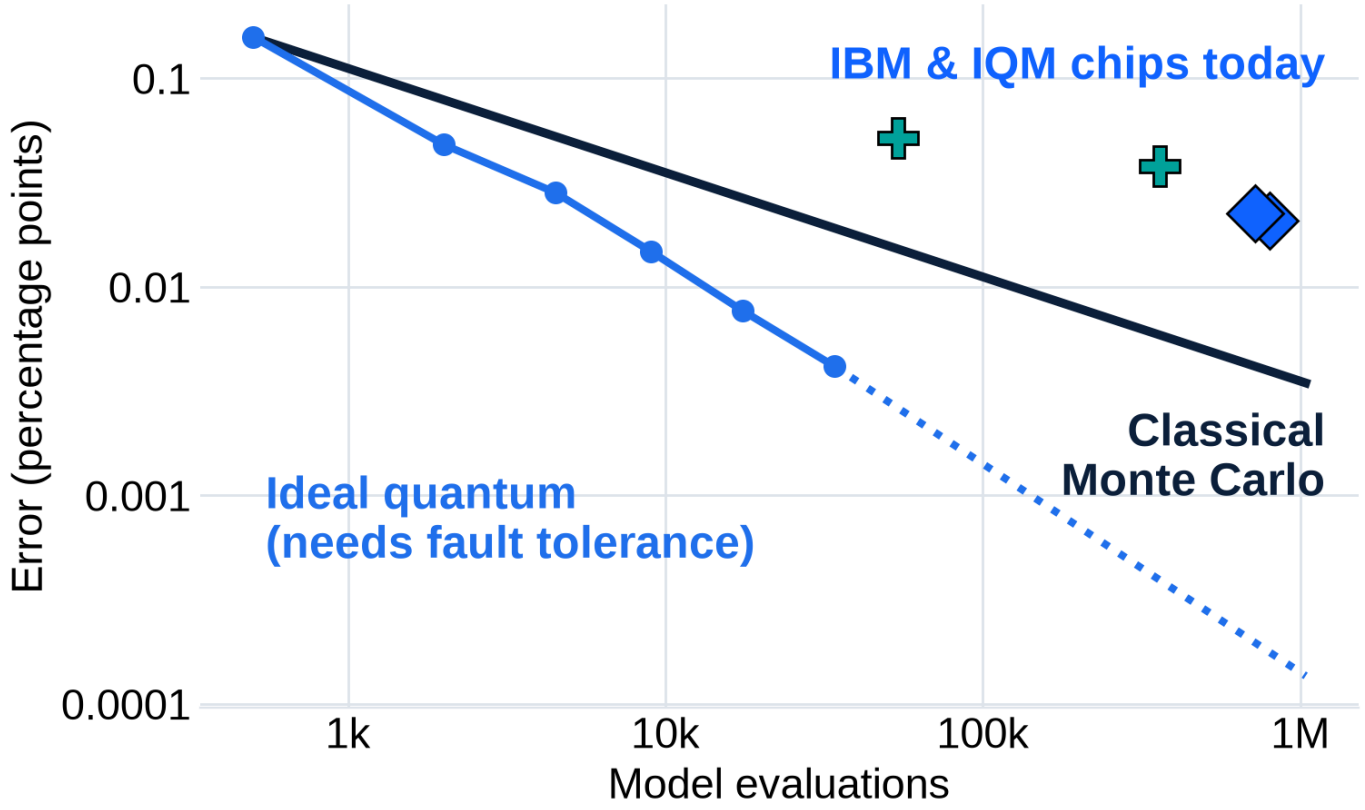
Exact	**0.125 %**
IBM	**0.149 %** 90 % CI 0.140–0.157
IQM	**0.164 %** 90 % CI 0.147–0.180

Calibrated; raw readings were 2–3× off

IQM Emerald (snow: 10 copies, 240k shots per level; 14 areas: job 01a11cd4…); IBM Marrakesh, IQM Garnet; bootstrap 90 % CI

QUANTUM, HONESTLY

Classical Monte Carlo still wins today



Kittilä and Ivalo both pass 1-in-50: error vs effort

Method ready, hardware not yet

Measured today: ~18 qubits, ~50 two-qubit gates work. Rare events (1-in-50) drown in noise on the 14-area loader. Noise also fattens tails: an independent control run reads 2× high.

Prices are computed classically; quantum is the R&D track.

Advantage needs fault-tolerant scale (cf. derivative pricing: ~8k logical qubits, Chakrabarti 2021)

Source: Tulva simulations (qae-sim) + IBM Marrakesh / IQM Garnet / IQM Emerald runs; Chakrabarti et al., Quantum 5, 463 (2021)

BUSINESS

Who buys, why, and why now

WHO BUYS

Insurers & reinsurers

HOW WE EARN

Index + pricing engine, fee per deal (€25k pilot)

WHY NOW

Solvency II flood capital, Jan 2027 — snowmelt correlation set to 0

SCALE

Snowmelt regions: Sweden, Norway, Alps, Canada

Ask **One insurer's flood claims for 8 weeks → basis-risk map + firm quote for 1 Jan 2028.**

Sources: Del. Reg. (EU) 2026/269; EIOPA-BoS-24/462 §4.70 (25 % snowmelt correlation rejected)

Tulva

Next March, the snow will warn us again.

This time, it will have a price.

BUILT THIS WEEKEND

SYKE data pipeline | Joint flood model | Index pricing | Live app | QAE on IBM & IQM

RESOURCES USED

SYKE open data | IBM Marrakesh | IQM Garnet & Emerald (Resonance) | Qrisp | Qiskit | AI coding agents

Toivo Kallio · **Vaishnav Rane**

Team 21 STUHI · Quantum x Finance 2026, Hanken × Ultrahack